

Solved selected problems of Classical Electrodynamics - Hans Ohanian

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Chapter 7 - Vector Calculus in Spacetime

Problems

Solution 2.

- (a) Let $A(\mu, \nu)$ be an object with two indices and suppose that $A(\mu, \nu)B^\mu$ is a vector when B^μ is a vector.

Let us consider the transformation law of the vector $A(\mu, \nu)B^\mu$

$$A'(\mu, \nu)B'^\mu = a_\nu^\alpha A(\mu, \alpha)B^\mu$$

Since B^μ is also a vector then we know that

$$B^\mu = a_\sigma^\mu B'^\sigma$$

Hence

$$A'(\mu, \nu)B'^\mu = a_\nu^\alpha A(\mu, \alpha)B^\mu = a_\sigma^\mu a_\nu^\alpha A(\mu, \alpha)B'^\sigma$$

Renaming the dummy indices on the RHS $\sigma \leftrightarrow \mu$ we get that

$$A'(\mu, \nu)B'^\mu = a_\mu^\sigma a_\nu^\alpha A(\sigma, \alpha)B'^\mu$$

Since B'^μ is a vector, it can be the vector $B'^\mu = (1, 0, 0, 0)$ and hence the equation becomes

$$A'(t, \nu) = a_t^\sigma a_\nu^\alpha A(\sigma, \alpha)$$

For the vector $B'^\mu = (0, 1, 0, 0)$ we get that

$$A'(x, \nu) = a_x^\sigma a_\nu^\alpha A(\sigma, \alpha)$$

The same can be proven for the vectors $B'^\mu = (0, 0, 1, 0)$ and $B'^\mu = (0, 0, 0, 1)$.

With this selection of vectors we see that $A(\mu, \nu)$ transforms like a tensor, and since this must hold for every vector, then it must be that $A(\mu, \nu)$ is a tensor.

- (b) Let $A(\mu, \nu)$ be an object with two indices and suppose that $A(\mu, \nu)C^{\mu\nu}$ is a scalar when $C^{\mu\nu}$ is a tensor. Scalars are invariant under transformation laws so

$$A'(\mu, \nu)C'^{\mu\nu} = A(\mu, \nu)C^{\mu\nu}$$

Since $C^{\mu\nu}$ is a tensor then we know that

$$C^{\mu\nu} = a_{\alpha}^{\mu} a_{\beta}^{\nu} C'^{\alpha\beta}$$

Then

$$A'(\mu, \nu)C'^{\mu\nu} = a_{\alpha}^{\mu} a_{\beta}^{\nu} A(\mu, \nu)C'^{\alpha\beta}$$

Renaming the dummy indices on the RHS $\alpha \leftrightarrow \mu$ and $\beta \leftrightarrow \nu$, we get that

$$A'(\mu, \nu)C'^{\mu\nu} = a_{\mu}^{\alpha} a_{\nu}^{\beta} A(\alpha, \beta)C'^{\mu\nu}$$

As in part (a), if we select

$$C'^{\mu\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Then the equation becomes

$$A'(t, t) = a_t^{\alpha} a_t^{\beta} A(\alpha, \beta)$$

The same can be shown for every component of $A(\mu, \nu)$, setting to 1 only the component we are interested in, in the tensor $C'^{\mu\nu}$.

With this selection of tensors we see that $A(\mu, \nu)$ transforms like a tensor, and since this must hold for every tensor, then it must be that $A(\mu, \nu)$ is a tensor.

□

Solution 3. Let A^μ and B_μ be two vectors such that $A^\mu B_\mu = 0$

- (a) Suppose $A^\mu A_\mu > 0$. We note that we can normalize $A^\mu A_\mu$ such that $A^\mu A_\mu = 1$, so we may take A^μ with three other vectors x^μ, y^μ and z^μ as a orthonormal basis.

Let us suppose A^μ is the timelike vector and the other three are space-like vectors.

Since $A^\mu B_\mu = 0$ then must be that B^μ is orthogonal to A^μ and hence it must be a linear combination of the other three vectors i.e.

$$B^\mu = ax^\mu + by^\mu + cz^\mu$$

Where a, b, c are scalars. Then

$$\begin{aligned} B^\mu B_\mu &= -(a^2 x^1 x_1 + b^2 y^1 y_1 + c^2 z^1 z_1) - (a^2 x^2 x_2 + b^2 y^2 y_2 + c^2 z^2 z_2) \\ &\quad - (a^2 x^3 x_3 + b^2 y^3 y_3 + c^2 z^3 z_3) \\ &= -(a^2 + b^2 + c^2) - (a^2 + b^2 + c^2) - (a^2 + b^2 + c^2) \end{aligned}$$

Where we used that $x^i y_i = x^i z_i = y^i z_i = 0$ and that $x^i x_i = y^i y_i = z^i z_i = 1$ for $i \in \{1, 2, 3\}$. Therefore $B^\mu B_\mu < 0$.

- (b) Suppose $A^\mu A_\mu = 0$ and let us suppose $B^\mu B_\mu > 0$ we want to arrive at a contradiction.

Let $B^\mu = (1, 0, 0, 0)$ since $A^\mu B_\mu = 0$ we get that $A^0 = 0$, also since $A^\mu A_\mu = 0$ we get that

$$\begin{aligned}(A^0)^2 - (A^1)^2 - (A^2)^2 - (A^3)^2 &= 0 \\ (A^1)^2 + (A^2)^2 + (A^3)^2 &= 0\end{aligned}$$

This is a contradiction since it cannot be unless $A^\mu = (0, 0, 0, 0)$ which is a trivial solution and for which B^μ can be any vector.

Therefore must be that $B^\mu B_\mu \leq 0$.

If $B^\mu B_\mu = 0$ then B^μ is a lightlike vector as A^μ .

Let us take a reference frame such that $A^\mu = (1, 1, 0, 0)$.

Since we know that $A^\mu B_\mu = A'^\mu B'_\mu$ when A'^μ and B'_μ are in a different reference frame, then taking B^μ in the same reference frame where $A^\mu = (1, 1, 0, 0)$, must be that

$$\begin{aligned}A^\mu B_\mu &= 0 \\ A^0 B^0 - A^1 B^1 &= 0 \\ B^0 - B^1 &= 0\end{aligned}$$

Then $B^0 = B^1$, but also we know that $B^\mu B_\mu = 0$ so

$$\begin{aligned}(B^0)^2 - (B^1)^2 - (B^2)^2 - (B^3)^2 &= 0 \\ -(B^2)^2 - (B^3)^2 &= 0\end{aligned}$$

Which implies that in this frame B^μ must be of the form

$$B^\mu = (B^0, B^0, 0, 0)$$

Therefore must be that B^μ is porportional to A^μ i.e. $B^\mu = kA^\mu$.

□

Solution 7.

- (a) We know that the relativistic momentum is given by

$$\mathbf{p} = \frac{m\mathbf{v}}{\sqrt{1 - v^2/c^2}}$$

Since the acceleration and the velocity are always parallel then we may define our coordinate system such that

$$p_x = \frac{mv_x}{\sqrt{1 - v_x^2/c^2}}$$

Taking the derivative with respect to time, we get that

$$\begin{aligned} \frac{dp_x}{dt} &= \frac{mv_x^2}{c^2(1 - v_x^2/c^2)^{3/2}} \frac{dv_x}{dt} + \frac{m}{\sqrt{1 - v_x^2/c^2}} \frac{dv_x}{dt} \\ &= \frac{m}{(1 - v_x^2/c^2)^{3/2}} \frac{dv_x}{dt} \left(\frac{v_x^2}{c^2} + \frac{(1 - v_x^2/c^2)^{3/2}}{\sqrt{1 - v_x^2/c^2}} \right) \\ &= \frac{m}{(1 - v_x^2/c^2)^{3/2}} \frac{dv_x}{dt} \left(\frac{v_x^2}{c^2} + 1 - \frac{v_x^2}{c^2} \right) \\ &= \frac{m}{(1 - v_x^2/c^2)^{3/2}} \frac{dv_x}{dt} \end{aligned}$$

Since when the acceleration and the velocity are parallel we can always select the coordinate system such that they are in the same axis (in this case the x axis) then we can write that

$$\frac{d\mathbf{p}}{dt} = \frac{m}{(1 - v^2/c^2)^{3/2}} \frac{d\mathbf{v}}{dt}$$

Therefore, comparing with the equation $d\mathbf{p}/dt = m(v)d\mathbf{v}/dt$ we see that

$$m(v) = \frac{m}{(1 - v^2/c^2)^{3/2}}$$

- (b) Let us consider now a particle in uniform circular motion such that the acceleration and the velocity are always perpendicular. Given that the magnitude of the velocity is constant in a uniform circular motion when taking the derivative with respect to time of

$$\mathbf{p} = \frac{m\mathbf{v}}{\sqrt{1 - v^2/c^2}}$$

We get that

$$\frac{d\mathbf{p}}{dt} = \frac{m}{\sqrt{1 - v^2/c^2}} \frac{d\mathbf{v}}{dt}$$

Therefore $m(v)$ must be

$$m(v) = \frac{m}{\sqrt{1 - v^2/c^2}}$$

- (c) Finally, let us consider the case where the acceleration is neither parallel nor perpendicular to the velocity.

Taking the derivative with respect to time of the relativistic momentum where $\mathbf{v}$ points in an arbitrary direction, we get that

$$\frac{d\mathbf{p}}{dt} = m \frac{d}{dt} \left(\frac{1}{\sqrt{1 - v^2/c^2}} \right) \mathbf{v} + \frac{m}{\sqrt{1 - v^2/c^2}} \frac{d\mathbf{v}}{dt}$$

We note that

$$\begin{aligned} \frac{d}{dt} \left(\frac{1}{\sqrt{1 - (\mathbf{v} \cdot \mathbf{v})/c^2}} \right) &= -\frac{1}{2} \left(1 - \frac{v^2}{c^2} \right)^{-3/2} \frac{d}{dt} \left(1 - \frac{\mathbf{v} \cdot \mathbf{v}}{c^2} \right) \\ &= \frac{1}{2} \left(1 - \frac{v^2}{c^2} \right)^{-3/2} \frac{2}{c^2} \left(\mathbf{v} \cdot \frac{d\mathbf{v}}{dt} \right) \\ &= \frac{\mathbf{v} \cdot \frac{d\mathbf{v}}{dt}}{c^2 (1 - v^2/c^2)^{3/2}} \end{aligned}$$

So

$$\frac{d\mathbf{p}}{dt} = \frac{m \mathbf{v} \cdot \frac{d\mathbf{v}}{dt}}{c^2 (1 - v^2/c^2)^{3/2}} \mathbf{v} + \frac{m}{\sqrt{1 - v^2/c^2}} \frac{d\mathbf{v}}{dt}$$

Therefore, $d\mathbf{p}/dt$ is not going to have the direction of the velocity nor the direction of the acceleration, but a combination of both and hence there is no scalar factor $m(v)$ we can come up with such that

$$\frac{d\mathbf{p}}{dt} = m(v) \frac{d\mathbf{v}}{dt}$$

□

Solution 8. Since the proton is at rest, then in momentum space the vector representing the proton is vertical and points to the vertex of the mass shell (or the hyperboloid), this is the point $(p^0, 0, 0, 0)$.

If we assume the tachyon with $v = \infty$ is hitting the proton coming in the direction of the p^x -axis then the final state of the proton must be a vector with coordinates $(p^0, p^x, 0, 0)$ where $p^x \neq 0$, but we see that there is no point of the hyperboloid that matches this constraint.

Therefore, the proton cannot absorb this tachyon. □

Solution 9.

- (a) Let a extremely relativistic particle where $v \approx c$. This particle has an energy of

$$E = \frac{mc^2}{\sqrt{1 - v^2/c^2}}$$

Given that $v \approx c$ we can define a small quantity $\epsilon = 1 - v/c$ such that

$$\frac{v^2}{c^2} = (\epsilon - 1)^2 = \epsilon^2 - 2\epsilon + 1 \approx -2\epsilon + 1$$

So

$$1 - \frac{v^2}{c^2} = 1 + 2\epsilon - 1 = 2\epsilon$$

Therefore

$$\begin{aligned} E &= \frac{mc^2}{\sqrt{2\epsilon}} \\ \epsilon &= \frac{1}{2} \left(\frac{mc^2}{E} \right)^2 \\ \frac{1}{c}(c - v) &= \frac{1}{2} \left(\frac{mc^2}{E} \right)^2 \\ c - v &= \frac{c}{2} \left(\frac{mc^2}{E} \right)^2 \end{aligned}$$

- (b) Given a proton of 1000 *GeV* the velocity difference is

$$c - v = \frac{299 \times 10^6 \text{ m/s}}{2} \left(\frac{938.27 \text{ MeV}}{1 \times 10^6 \text{ MeV}} \right)^2 = 131.6 \text{ m/s}$$

- (c) Given an electron of 40 *GeV* the velocity difference is

$$c - v = \frac{299 \times 10^6 \text{ m/s}}{2} \left(\frac{0.511 \text{ MeV}}{4 \times 10^4 \text{ MeV}} \right)^2 = 0.0243 \text{ m/s}$$

□

Solution 10. Let us consider a particle of mass $m_1 = 1 \text{ g}$ moving in the positive x direction with a velocity of $v_1 = 0.9$ and another particle of mass $m_2 = 1 \text{ g}$ moving in the negative x direction with velocity $v_2 = -0.99$. We assume $c = 1$. Then the total relativistic momentum of the system is

$$\begin{aligned} P &= \frac{m_1 v_1}{\sqrt{1 - v_1^2}} + \frac{m_2 v_2}{\sqrt{1 - v_2^2}} \\ &= 2.06 - 7.01 \\ &= -4.94 \end{aligned}$$

Also, the total energy of the system is

$$\begin{aligned} E &= \frac{m_1}{\sqrt{1 - v_1^2}} + \frac{m_2}{\sqrt{1 - v_2^2}} \\ &= 2.29 + 7.08 \\ &= 9.37 \end{aligned}$$

So the velocity of the center of momentum from the laboratory frame is

$$v_{C.M.} = \frac{P}{E} = \frac{-4.94}{9.37} = -0.52$$

On the other hand, the velocity of the center of mass is

$$v_{C.mass} = \frac{m_1 v_1 + m_2 v_2}{m_1 + m_2} = \frac{0.9 - 0.99}{2} = -0.045$$

Therefore we see that for relativistic particles $v_{C.M.} \neq v_{C.mass}$.

□

Solution 20.

- (a) From the conservation of energy and momentum in the instantaneous rest frame $x'y'z't'$ of the rocket we have that

$$dv' = \frac{v_{ex}}{M} \frac{dm}{\sqrt{1 - v_{ex}^2/c^2}} = -\frac{v_{ex}}{M} dM$$

Also, from the velocity-combination formula we can approximate it as

$$\begin{aligned} v + dv &= \frac{v + dv'}{1 + v dv'/c^2} \\ &\approx (v + dv') \left(1 - \frac{v dv'}{c^2} \right) \\ &\approx v + dv' - \frac{v^2 dv'}{c^2} \\ &= v + \left(1 - \frac{v^2}{c^2} \right) dv' \end{aligned}$$

Where we used the binomial approximation and we dropped the dv'^2 term. Then

$$dv = \left(1 - \frac{v^2}{c^2} \right) dv'$$

Replacing dv' and integrating we get that

$$\begin{aligned} \frac{dv}{1 - v^2/c^2} &= -\frac{v_{ex}}{M} dM \\ \int_0^v \frac{dv}{1 - v^2/c^2} &= -v_{ex} \int_{M_i}^M \frac{dM}{M} \\ c \arctan\left(\frac{v}{c}\right) &= -v_{ex} [\log(M) - \log(M_i)] \\ \frac{c}{v_{ex}} \arctan\left(\frac{v}{c}\right) &= \log\left(\frac{M_i}{M}\right) \\ \frac{c}{2v_{ex}} \log\left(\frac{1 + v/c}{1 - v/c}\right) &= \log\left(\frac{M_i}{M}\right) \end{aligned}$$

Therefore

$$\begin{aligned} \log\left(\frac{M_i}{M}\right) &= \log\left(\left(\frac{c + v}{c - v}\right)^{\frac{c}{2v_{ex}}}\right) \\ \frac{M_i}{M} &= \left(\frac{c + v}{c - v}\right)^{\frac{c}{2v_{ex}}} \end{aligned}$$

- (b) Let $v_{ex} = 3 \times 10^5 \text{ cm/s}$, $v = 0.1c$ and $M = 1 \text{ ton}$ then the mass of fuel needed is

$$\begin{aligned}
 M_i - M &= M \left(\frac{c+v}{c-v} \right)^{\frac{c}{2v_{ex}}} - M \\
 &= \left(\frac{1.1}{0.9} \right)^{\frac{3 \times 10^{10}}{6 \times 10^5}} - 1 \\
 &= 3.12 \times 10^{4354} \text{ ton}
 \end{aligned}$$

- (c) Let $v_{ex} = c$, $v = 0.1c$ and $M = 1 \text{ ton}$ then the mass of fuel needed is

$$\begin{aligned}
 M_i - M &= M \left(\frac{c+v}{c-v} \right)^{\frac{c}{2v_{ex}}} - M \\
 &= \left(\frac{1.1}{0.9} \right)^{1/2} - 1 \\
 &= 0.105 \text{ ton}
 \end{aligned}$$

□

Solution 22. Let $\Phi(t, \mathbf{x})$ be a scalar field. To show that

$$f(x^\mu) = \int \Phi(x^\mu) dV dt$$

is a scalar, we must show that f is invariant under Lorentz transformations.

If $f(x^\mu)$ is a scalar then must be that $f(x^\mu) = f(a_\alpha^\mu x'^\alpha)$.

Let us compute $f(a_\alpha^\mu x'^\alpha)$ as follows

$$\begin{aligned} f(a_\alpha^\mu x'^\alpha) &= \int \Phi(a_\alpha^\mu x'^\alpha) dV' dt' \\ &= \int \Phi(x^\mu) \sqrt{1 - V^2} dV \frac{dt}{\sqrt{1 - V^2}} \\ &= \int \Phi(x^\mu) dV dt \\ &= f(x^\mu) \end{aligned}$$

We used that Φ is a scalar field and hence $\Phi(a_\alpha^\mu x'^\alpha) = \Phi(x^\mu)$, also, we used the volume contraction and time dilation equations. $\square$

Solution 23. Let the current density j^μ differ from zero only in a volume of finite extent in the x, y, z directions.

Let us integrate $\partial_\mu j^\mu = 0$ over the volume of a four-dimensional world tube that encloses the region of interest and extends from $x^0 = \text{constant}$ to $x'^0 = \text{constant}$.

$$\int_{\Omega} \partial_\mu j^\mu d\Omega = 0$$

We note that we are integrating over a four-dimensional volume Ω , then applying Gauss' theorem we get that

$$\int_V j^\mu n_\mu dV = 0$$

Where V is the surface of our four-dimensional volume. Given that j^μ is only non-zero in a finite volume which is enclosed by the integral, then the "flux" is only non-zero in the direction of the axis x^0 , so the integral becomes

$$\int_V j'^0 dV - \int_V j^0 dV = 0$$

Therefore

$$Q' = \int_V j'^0 dV = \int_V j^0 dV = Q$$

This implies that the total charge is a scalar under Lorentz transformations.

□

Solution 24. In the previous problem we found that

$$Q = \int_V J^0 dV$$

Since J^μ is different from zero only in a volume of a finite extent then the exchange of charge with the space outside the volume is zero, then must be that the total charge Q held inside the volume doesn't change with time, therefore

$$\frac{dQ}{dt} = \frac{d}{dt} \left(\int_V J^0 dV \right) = 0$$

□

Solution 25. In Exercise 17 we showed that an electrically neutral wire carrying a current in the xyz reference frame is not electrically neutral in the reference frame $x'y'z'$ and the charge density (in this case a linear charge density) is given by

$$\lambda' = -\frac{V j_x / c^2}{\sqrt{1 - V^2/c^2}}$$

So if $j_x = 3 \times 10^{11} \text{ esu/s}$ and $V = 0.8c$ we get that the charge per unit of length is

$$\lambda' = -\frac{0.8 \cdot 3 \times 10^{11} / 3 \times 10^{10}}{\sqrt{1 - 0.8^2}} = -13.33 \text{ esu/cm}$$

□

Solution 26. We know that an electrically neutral wire carrying a current in the xyz reference frame is not electrically neutral in the reference frame $x'y'z'$ and the charge density (in this case a linear charge density) is given by

$$\lambda' = -\frac{VI_x/c^2}{\sqrt{1-V^2/c^2}}$$

Where I_x is the x component of the current I , in this case I_x is related to θ by

$$I_x = I \cos \theta$$

Therefore the charge density observed in the reference frame $x'y'z'$ is

$$\lambda' = -\frac{VI \cos \theta/c^2}{\sqrt{1-V^2/c^2}}$$

□

Solution 27. Let us consider two large, horizontal, parallel conducting plates carrying currents in opposite directions, the positive x direction and the negative x direction. The current is uniformly distributed over each plate, with k esu of charge per second flowing across each centimeter measured along the plate perpendicularly to the current.

An observer in a reference frame moving at speed V in the positive x direction will see a charge density and a current density of

$$\sigma'_{+x} = -\frac{Vk}{c^2\sqrt{1-V^2/c^2}} \quad j'_{+x} = \frac{k}{\sqrt{1-V^2/c^2}}$$

for the plate with current in the positive x direction.

For the plate with current in the negative x direction, the observer will experience a charge density and a current density of

$$\sigma'_{-x} = \frac{Vk}{c^2\sqrt{1-V^2/c^2}} \quad j'_{-x} = -\frac{k}{\sqrt{1-V^2/c^2}}$$

We know that the electric field due to two charged plates is zero outside the plates (they cancel out) and in between them is

$$E_z = 4\pi\sigma$$

Therefore

$$E_z = \frac{4\pi Vk}{c^2\sqrt{1-V^2/c^2}}$$

□